MODELING THE EVOLUTION OF TWITCH VIEWERSHIP DURING THE COVID-19 PANDEMIC

BY KIRA DAMO

TWITCH AND THE PANDEMIC

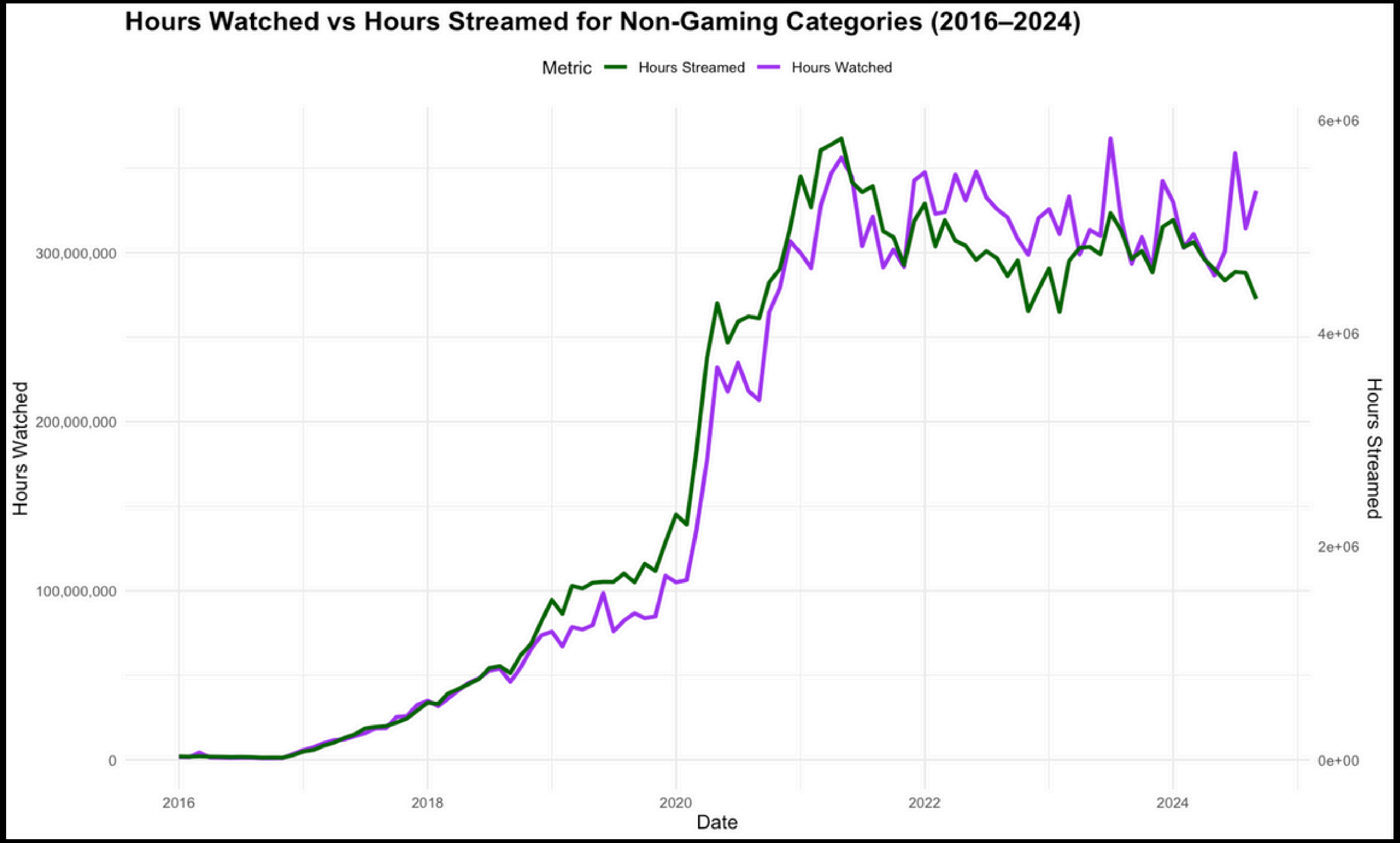
- Twitch is a livestreaming platform that is widely known for gaming content.
- Users with the necessary software and equipment can broadcast content, referred to as streamers, and other users can interact with the content as well.
- Mandatory lockdowns and the enforcement of social distancing left people with a yearning for community and entertainment.
- For many people, Twitch was the most optimal way to cure the withdrawals of social physical interactions.
- **This project will explore the changes in the top categories during the pandemic and how that correlates to current events and cultural shifts offline**

THE DATA

- The dataset consists of the 200 top streamed categories from January of 2016 to September of 2024.
- The variables are Rank, Game, Month, Year, Hours Watched, Hours Streamed, Peak Viewers, Peak Channels, Streamers, Average Viewers, Average Channels, and Average Viewer Ratio
- In total there are 2,360 categories that ranked in the top 200 from 2016 to 2024.

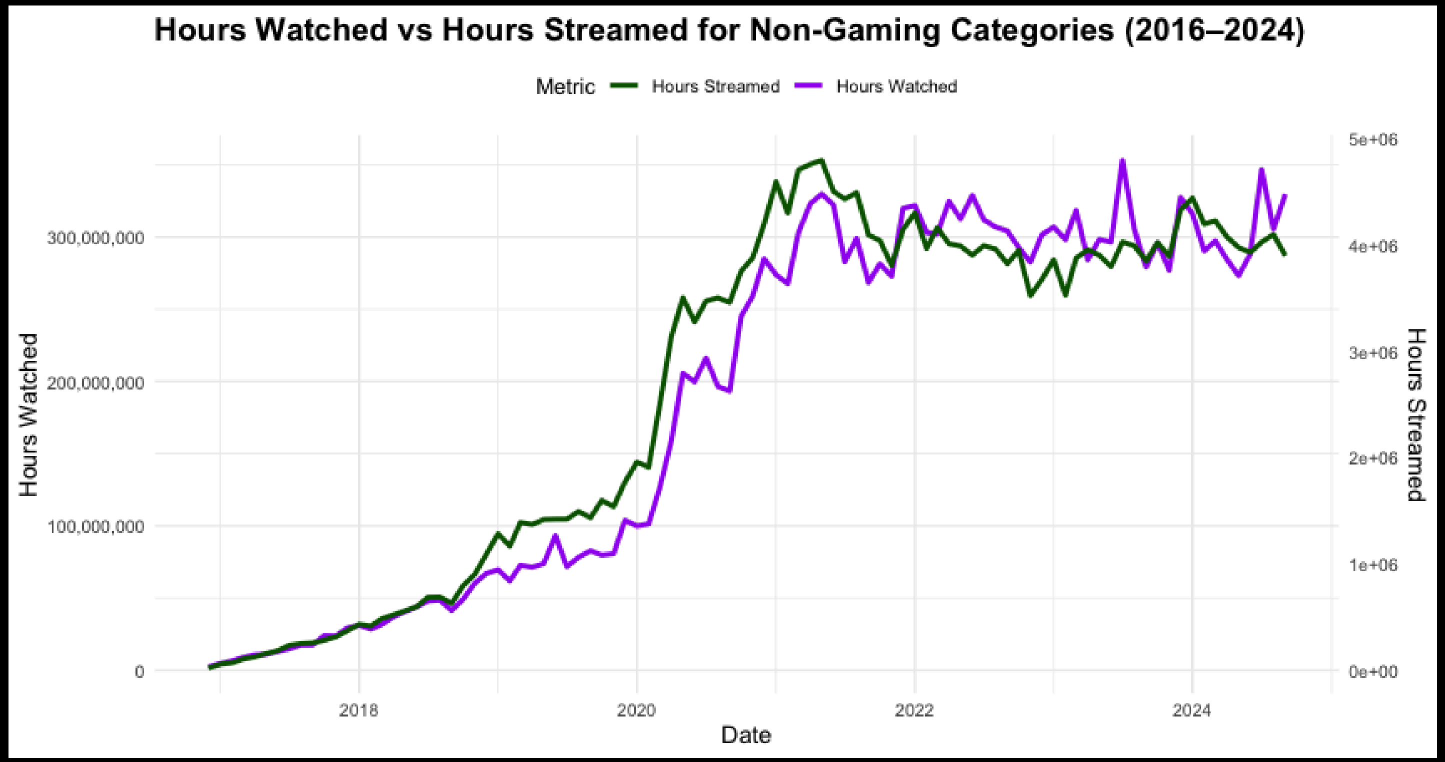
VIEWERSHIP THROUGHOUT THE PANDEMIC

- According to the CDC Museum COVID-19 Timeline, COVID-19 was officially identified as a pandemic on March 20th, 2020 by the World Health Organization.
- As demonstrated by the graph, viewership and streamed hours rapidly increased after 2020 and started to plateau around 2023 which is when the WHO concluded COVID-19 as an international health concern.



HOW THE PANDEMIC CHANGED THE STREAMING LANDSCAPE

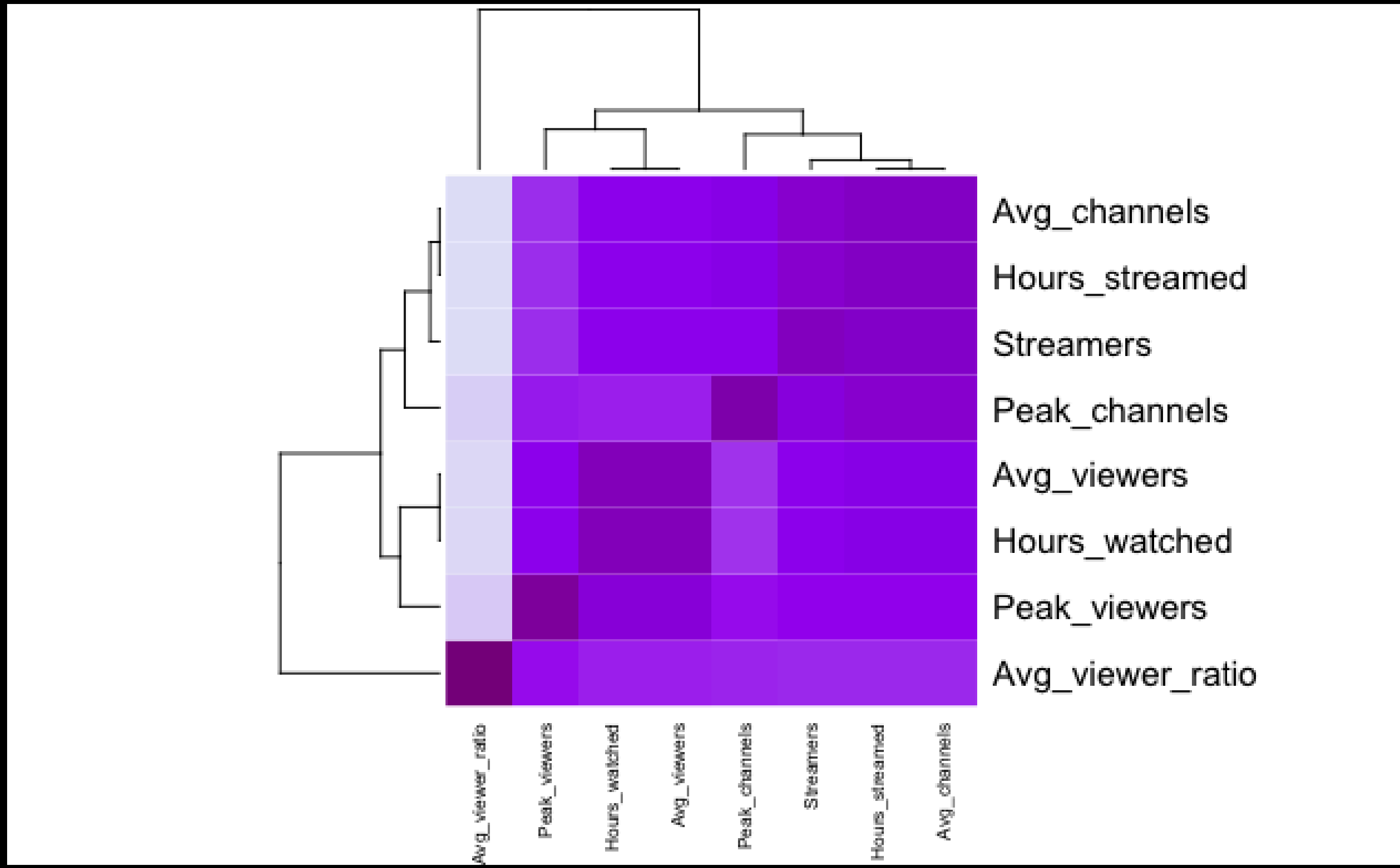
- Twitch is primarily a streaming platform for gaming content but the desire for close-knit communities changed the various forms of content being produced on the site.
- With the introduction of Just Chatting as a category in 2018 along with other non-gaming categories, viewership for these categories significantly increased during the pandemic.



Non-gaming categories considered:
ASMR, Art, Just Chatting, Food & Drink, Special Events, Makers & Crafting, Travel & Outdoors, Science & Technology, IRL, Politics, and Pools, Hot Tubs, and Beaches

FORMING A LINEAR REGRESSION MODEL

- To create a linear regression model that will optimally allow us to analyze the evolution of Twitch during the pandemic, we must narrow down the variables.
- Using this heatmap, we want to eliminate variables that are highly correlated to avoid redundancies and multicollinearity



Highest Correlated Pairs of Variables

	Variable1	Variable2	Correlation
1	Avg_viewers	Hours_watched	0.9995832
2	Avg_channels	Hours_streamed	0.9995719
3	Streamers	Hours_streamed	0.9322948
4	Avg_channels	Streamers	0.9321782
5	Avg_channels	Peak_channels	0.7831366
6	Peak_channels	Hours_streamed	0.7826466
7	Avg_channels	Avg_viewers	0.7360277
8	Hours_streamed	Hours_watched	0.7358847
9	Avg_channels	Hours_watched	0.7357373
10	Avg_viewers	Hours_streamed	0.7355292
11	Streamers	Peak_channels	0.7240835

FORMING A LINEAR REGRESSION MODEL

First Order Model:

- In order to assess which variables to keep and which are significant we must implement a first order model while also being mindful of the environment and the fluctuations that may occur.
- The numeric variables being kept and observed are Hours_Streamed, Hours_Watched, and Avg_Viewer_Ratio this way we can analyze stream activity, engagement, and how that curates the top categories streamed on Twitch.

FIRST ORDER MODEL: VARIABLE SELECTION

```
Call:
lm(formula = Hours_watched ~ Hours_streamed + Peak_viewers +
    Peak_channels + Streamers + Avg_viewers + Avg_channels +
    Avg_viewer_ratio, data = twitch)

Residuals:
      Min        1Q    Median        3Q       Max
-16841075    -13406      -323     12220   9616888

Coefficients:
                   Estimate Std. Error  t value Pr(>|t|)
(Intercept)      -9.889e+03  2.766e+03   -3.575 0.000351 ***
Hours_streamed    2.570e+01  1.497e-01  171.665  < 2e-16 ***
Peak_viewers      1.022e-01  2.354e-02    4.343 1.41e-05 ***
Peak_channels    -4.187e+00  1.618e+00   -2.587 0.009688 **
Streamers        -2.139e-01  9.827e-02   -2.176 0.029534 *
Avg_viewers       7.286e+02  1.684e-01 4327.447  < 2e-16 ***
Avg_channels     -1.871e+04  1.094e+02 -171.075  < 2e-16 ***
Avg_viewer_ratio -1.805e-01  6.089e+00   -0.030 0.976348
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 361500 on 20992 degrees of freedom
Multiple R-squared:  0.9997,	Adjusted R-squared:  0.9997
F-statistic: 8.663e+06 on 7 and 20992 DF,  p-value: < 2.2e-16
```

- In this model there are several variables that show as significant especially since the p-value is less than the default level of significance (alpha = 0.05).
- If we take into account the heatmap and correlation matrix that was shown previously, we find redundancy and multicollinearity in some of these quantitative variables.
- If all variables were to be kept, then this model would suffer from over-fitting.
- Peak_viewers and Avg_viewers are synonymous to Hours_watched and likewise for Peak_channels, Streamers, and Hour_streamed.
- The Avg_viewer ratio shows as insignificant but after removing the highly correlated and redundant variables, the ratio will become more relevant.

FIRST ORDER MODEL: VARIABLE SELECTION

```
Call:
lm(formula = Hours_watched ~ Hours_streamed + Avg_viewer_ratio,
    data = twitch)

Residuals:
       Min         1Q     Median         3Q        Max
-135858125   -1544417    -964265    -412824  228351084

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)      1.002e+06  9.736e+04  10.287  < 2e-16 ***
Hours_streamed   2.509e+01  1.593e-01 157.556  < 2e-16 ***
Avg_viewer_ratio 7.974e+02  2.207e+02   3.612 0.000304 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 13150000 on 20997 degrees of freedom
Multiple R-squared:  0.5418,	Adjusted R-squared:  0.5418
F-statistic: 1.241e+04 on 2 and 20997 DF,  p-value: < 2.2e-16
```

- The Hours_streamed and the Avg_viewer_ratio was retained due to their low correlation values in the heatmap and the correlation matrix.
- Hours_streamed is highly significant since the volume of hour_watched relies on how much content is streamed to be watched. This is also apparent in the p-value
- Although Avg_viewer ratio is not statistically significant, it is important to keep to account for engagement between streamer and viewers.
- The F-test indicates that this model predicts at least one contributor to the total hours watched on Twitch.

FORMING A LINEAR REGRESSION MODEL

Second Order Model:

- Considering the large boom in popularity of Twitch, a quadratic model may be a better fit for this model. The first order model’s adjusted R-squared value was 0.5418.
- To test if the data better adheres to the predictive values, interaction terms will be introduced.

SECOND ORDER MODEL: VARIABLE SELECTION

```
Call:
lm(formula = Hours_watched ~ poly(Hours_streamed, 2) + poly(Avg_viewer_ratio,
    2), data = twitch)

Residuals:
       Min         1Q     Median         3Q        Max
-105422554   -1045035     353438     797225  230809943

Coefficients:
                          Estimate Std. Error t value Pr(>|t|)
(Intercept)                5831853      84037  69.396  < 2e-16 ***
poly(Hours_streamed, 2)1  2077809876  12194556 170.388  < 2e-16 ***
poly(Hours_streamed, 2)2  -718256914  12191273 -58.916  < 2e-16 ***
poly(Avg_viewer_ratio, 2)1  72018824  12193663   5.906 3.55e-09 ***
poly(Avg_viewer_ratio, 2)2 -86457488  12192166  -7.091 1.37e-12 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 12180000 on 20995 degrees of freedom
Multiple R-squared:  0.6073,	Adjusted R-squared:  0.6072
F-statistic:  8115 on 4 and 20995 DF,  p-value: < 2.2e-16
```

- This is a quadratic model without interaction terms.
- In this model, the adjusted R-squared increased from 0.5148 to 0.6072 meaning this model accounts for more of the variance than the first order model did.
- The p-test shows that the quadratic terms are also both significant since the p-value is less than 0.05.
- The F-test indicates that this model can contribute to the prediction of hours watched on Twitch.

SECOND ORDER MODEL: VARIABLE SELECTION

```
Call:
lm(formula = Hours_watched ~ poly(Hours_streamed, 2) + poly(Avg_viewer_ratio,
    2) + poly(Hours_streamed, 1):poly(Avg_viewer_ratio, 1), data = twitch)

Residuals:
   Min     1Q Median     3Q    Max
-38451    -76     20     96  51991

Coefficients:
                                                Estimate Std. Error   t value
(Intercept)                                    1.458e+07  1.495e+01  9.753e+05
poly(Hours_streamed, 2)1                       6.340e+09  4.601e+03  1.378e+06
poly(Hours_streamed, 2)2                      -1.746e+04  1.901e+03 -9.184e+00
poly(Avg_viewer_ratio, 2)1                     1.133e+10  1.136e+04  9.971e+05
poly(Avg_viewer_ratio, 2)2                    -1.443e+02  1.764e+03 -8.200e-02
poly(Hours_streamed, 1):poly(Avg_viewer_ratio, 1)  4.928e+12  4.913e+06  1.003e+06
                                              Pr(>|t|)
(Intercept)                                     <2e-16 ***
poly(Hours_streamed, 2)1                        <2e-16 ***
poly(Hours_streamed, 2)2                        <2e-16 ***
poly(Avg_viewer_ratio, 2)1                      <2e-16 ***
poly(Avg_viewer_ratio, 2)2                       0.935
poly(Hours_streamed, 1):poly(Avg_viewer_ratio, 1)   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1759 on 20994 degrees of freedom
Multiple R-squared:      1,     Adjusted R-squared:      1
F-statistic: 5.122e+11 on 5 and 20994 DF,  p-value: < 2.2e-16
```

- This is a quadratic model that includes the interaction terms from the previous model.
- There is one interaction term that is deemed insignificant and would need to be neglected if chosen.
- Seeing that the R-squared value for this model is equal to 1, the model may suffer overfitting the data.
- Although this model passes both the F-test and p-test, the overfitting suggests that this is not a valid model for predicting the hours watched on Twitch.

MODEL COMPARISON

- The Akaike Information Criterion (AIC) can help guage which model is the best for predicting the hours watched.
- The AIC measures the goodness of the fit and the amount of parameters in the model.
- The AIC can be misleading in this case since Model 2 may have insignificant polynomial terms and Model 3 may suffer from overfitting.
- To combat this, we must perform stepAIC to eliminate insignificant terms.

	df <dbl>	AIC <dbl>
model1	9	597122.7
model2	6	744838.9
model3	7	373458.2

MODEL COMPARISON

- Using StepAIC, any unnecessary variables are pruned.

Model 2: Quadratic Model with No Interaction Terms

```
Start:  AIC=685241.5
Hours_watched ~ poly(Hours_streamed, 2) + poly(Avg_viewer_ratio,
    2)

                             Df  Sum of Sq        RSS
<none>                                     3.1137e+18
- poly(Avg_viewer_ratio, 2)  2 1.2601e+16 3.1263e+18
- poly(Hours_streamed, 2)    2 4.8133e+18 7.9271e+18
                                AIC
<none>                       685241
- poly(Avg_viewer_ratio, 2) 685322
- poly(Hours_streamed, 2)   704861

Call:
lm(formula = Hours_watched ~ poly(Hours_streamed, 2) + poly(Avg_viewer_ratio,
    2), data = twitch)

Coefficients:
               (Intercept)    poly(Hours_streamed, 2)1
                   5831853                  2077809876
  poly(Hours_streamed, 2)2  poly(Avg_viewer_ratio, 2)1
                -718256914                    72018824
poly(Avg_viewer_ratio, 2)2
                 -86457488
```

Model 3: Quadratic Model with Interaction Terms

```
Start:  AIC=313860.8
Hours_watched ~ poly(Hours_streamed, 2) + poly(Avg_viewer_ratio,
    2) + poly(Hours_streamed, 1):poly(Avg_viewer_ratio, 1)

                                                   Df  Sum of Sq        RSS
<none>                                                            6.4987e+10
- poly(Avg_viewer_ratio, 2)                         2 3.0850e+18 3.0850e+18
- poly(Hours_streamed, 1):poly(Avg_viewer_ratio, 1)  1 3.1137e+18 3.1137e+18
- poly(Hours_streamed, 2)                           2 6.6844e+18 6.6844e+18
                                                     AIC
<none>                                            313861
- poly(Avg_viewer_ratio, 2)                        685045
- poly(Hours_streamed, 1):poly(Avg_viewer_ratio, 1) 685241
- poly(Hours_streamed, 2)                          701283

Call:
lm(formula = Hours_watched ~ poly(Hours_streamed, 2) + poly(Avg_viewer_ratio,
    2) + poly(Hours_streamed, 1):poly(Avg_viewer_ratio, 1), data = twitch)

Coefficients:
                                      (Intercept)
                                        1.458e+07
                         poly(Hours_streamed, 2)1
                                        6.340e+09
                         poly(Hours_streamed, 2)2
                                       -1.746e+04
                       poly(Avg_viewer_ratio, 2)1
                                        1.133e+10
                       poly(Avg_viewer_ratio, 2)2
                                       -1.443e+02
poly(Hours_streamed, 1):poly(Avg_viewer_ratio, 1)
                                        4.928e+12
```

MODEL COMPARISON

- The results from StepAIC conclude that all variables in both models must be kept.
- Since Model 3 requires all terms to minimize AIC and the R-squared value is equal to exactly 1, Model 3 is not the best model to go forward with.
- Model 1's fit is linear but since Twitch’s rise during the pandemic was explosive, a quadratic fit will be better for predicting hours watched. Model 1's R-squared value is less than Model 2's.
- Therefore, Model 2 is the best fit going forward to predict the hours watched since it fits the data more accurately and is not a linear fit.

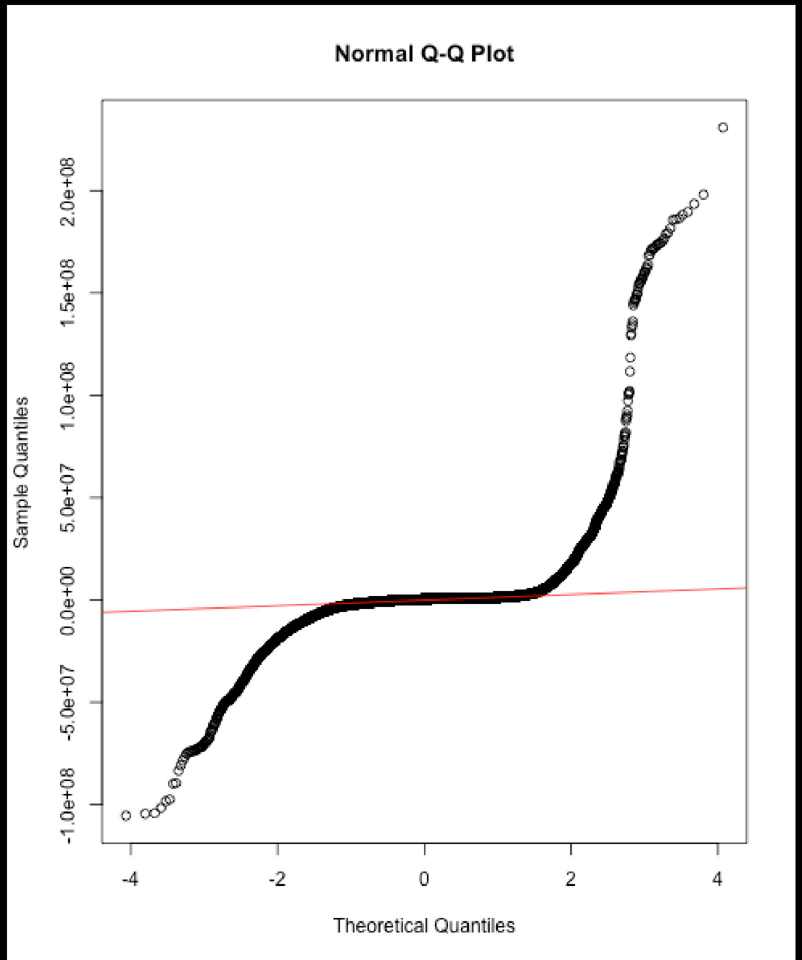
K-FOLD CROSS-VALIDATION

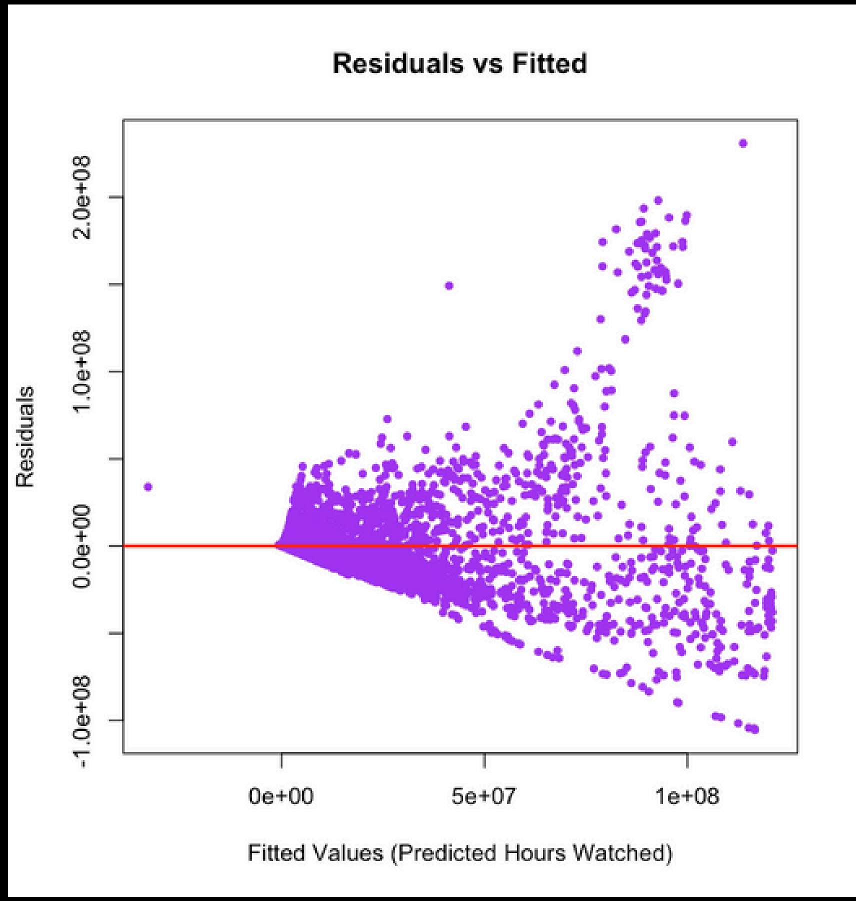
- To validate the effectiveness of the model on future data, cross-validation is performed.
- For this model, a 10-fold cross validation is performed.
- The cross validation shows that the model does not overfit or underfit and ~60% of the data is accounted for after training with 9 folds of the data and testing on one fold of data.

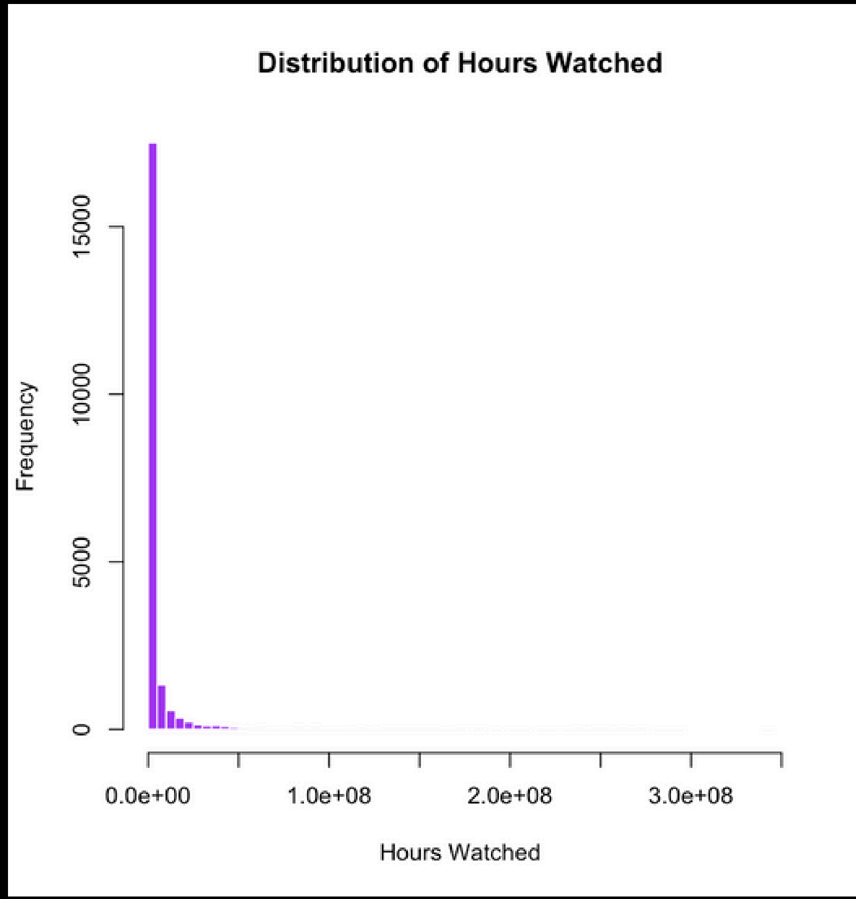
```
Linear Regression

21000 samples
    2 predictor

No pre-processing
Resampling: Cross-Validated (10 fold)
Summary of sample sizes: 18900, 18900, 18900, 18900, 18900, 18900, ...
Resampling results:

  RMSE      Rsquared   MAE
  12263308  0.6027475  3865818

Tuning parameter 'intercept' was held constant at a value of TRUE
```

RESIDUAL ANALYSIS

- Hours watched being a continuous variable allows the values to vary dramatically.
- Viewer engagement for a streamer can vary at any day which leads to larger residuals and causes heteroscedasticity.
- The funnel shape is not ideal for residual analysis since it indicates that homoscedasticity is violated.
- The distribution of Hours watched is right-skewed.
- These two graphs indicate that a log transformation may be a better model.

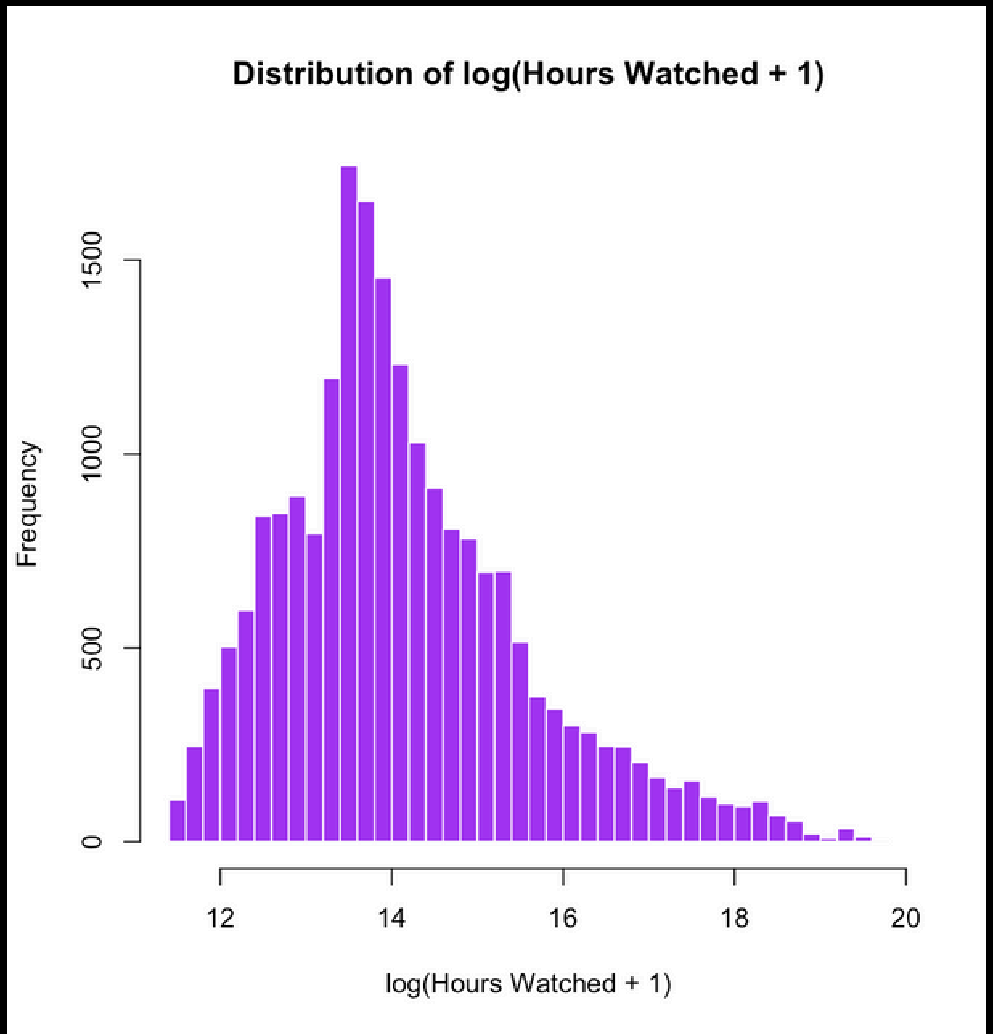


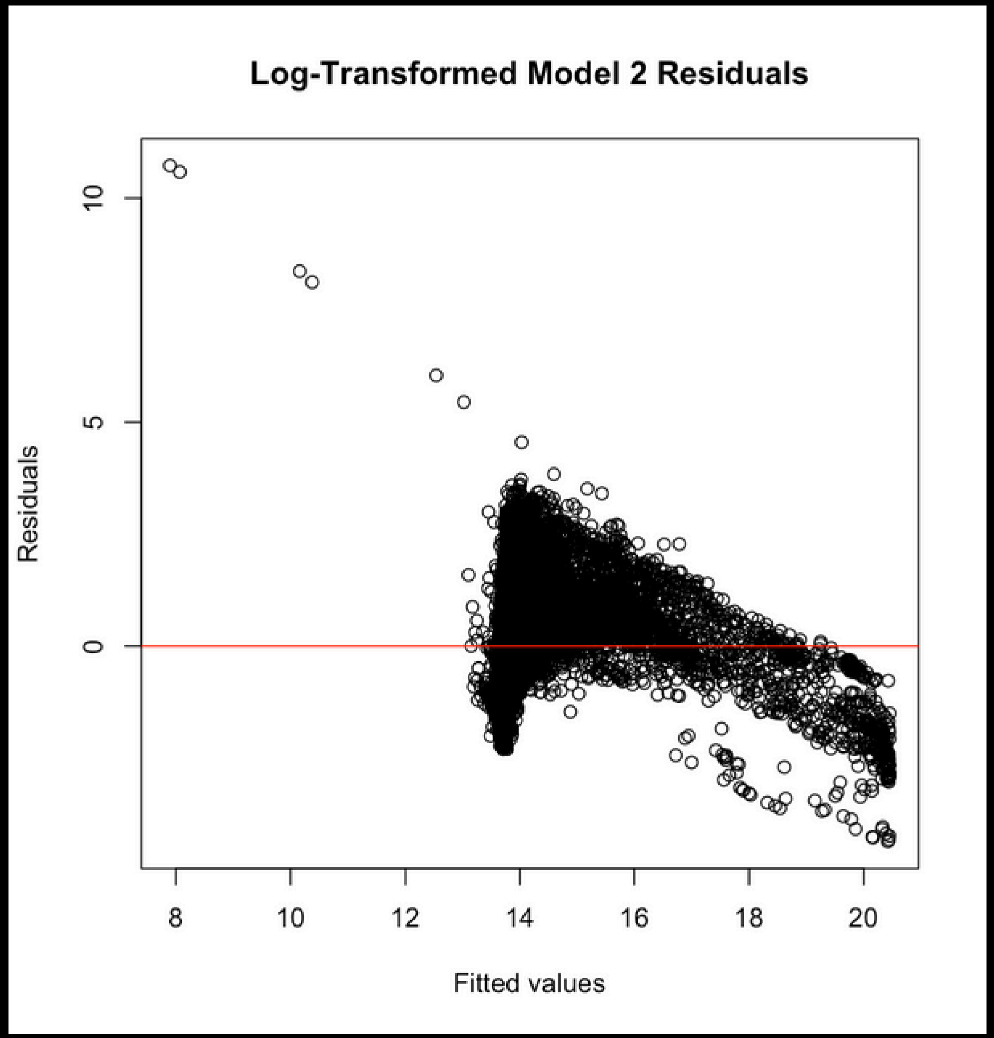


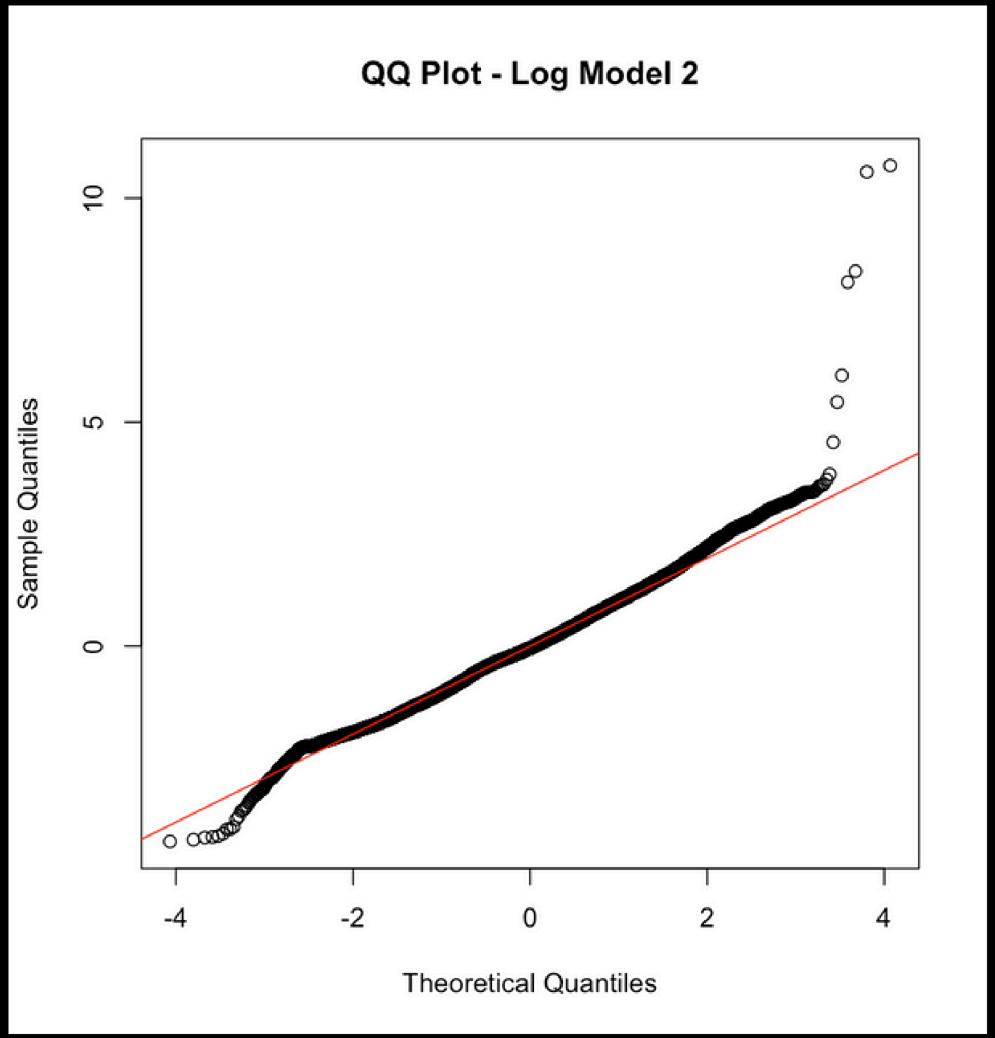


RESIDUAL ANALYSIS

- By doing a log transformation of the dependent variable, Hours Watched, The distribution is more symmetrical and the residuals are more randomly dispursed around 0.







FINAL MODEL

- To find the optimal model, variable selection process will need to be repeated.
- The Avg_view_ratio polynomial term’s p-value is greater than 0.05 and will need to be excluded.

```
Call:
lm(formula = log(Hours_watched + 1) ~ poly(Hours_streamed, 2) +
    poly(Avg_viewer_ratio, 2), data = twitch)

Residuals:
    Min      1Q  Median      3Q     Max
-4.3626 -0.6634 -0.0441  0.6617 10.7305

Coefficients:
                             Estimate Std. Error  t value Pr(>|t|)
(Intercept)                 14.147681   0.007117 1987.957   <2e-16 ***
poly(Hours_streamed, 2)1   123.595852   1.032696  119.683   <2e-16 ***
poly(Hours_streamed, 2)2   -78.827324   1.032418  -76.352   <2e-16 ***
poly(Avg_viewer_ratio, 2)1  -2.363759   1.032620   -2.289   0.0221 *
poly(Avg_viewer_ratio, 2)2   0.935902   1.032494    0.906   0.3647
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.031 on 20995 degrees of freedom
Multiple R-squared:  0.491,	Adjusted R-squared:  0.4909
F-statistic:  5064 on 4 and 20995 DF,  p-value: < 2.2e-16
```

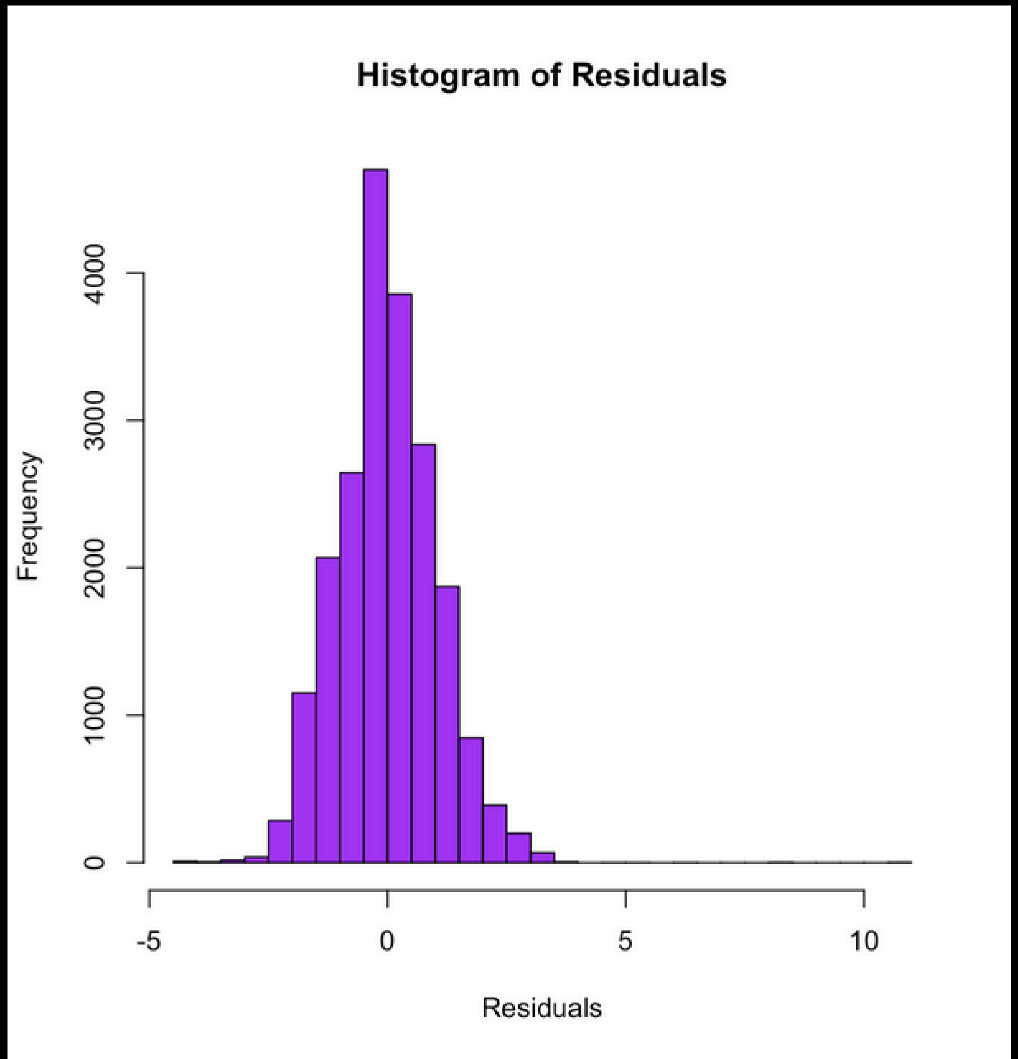
FINAL MODEL

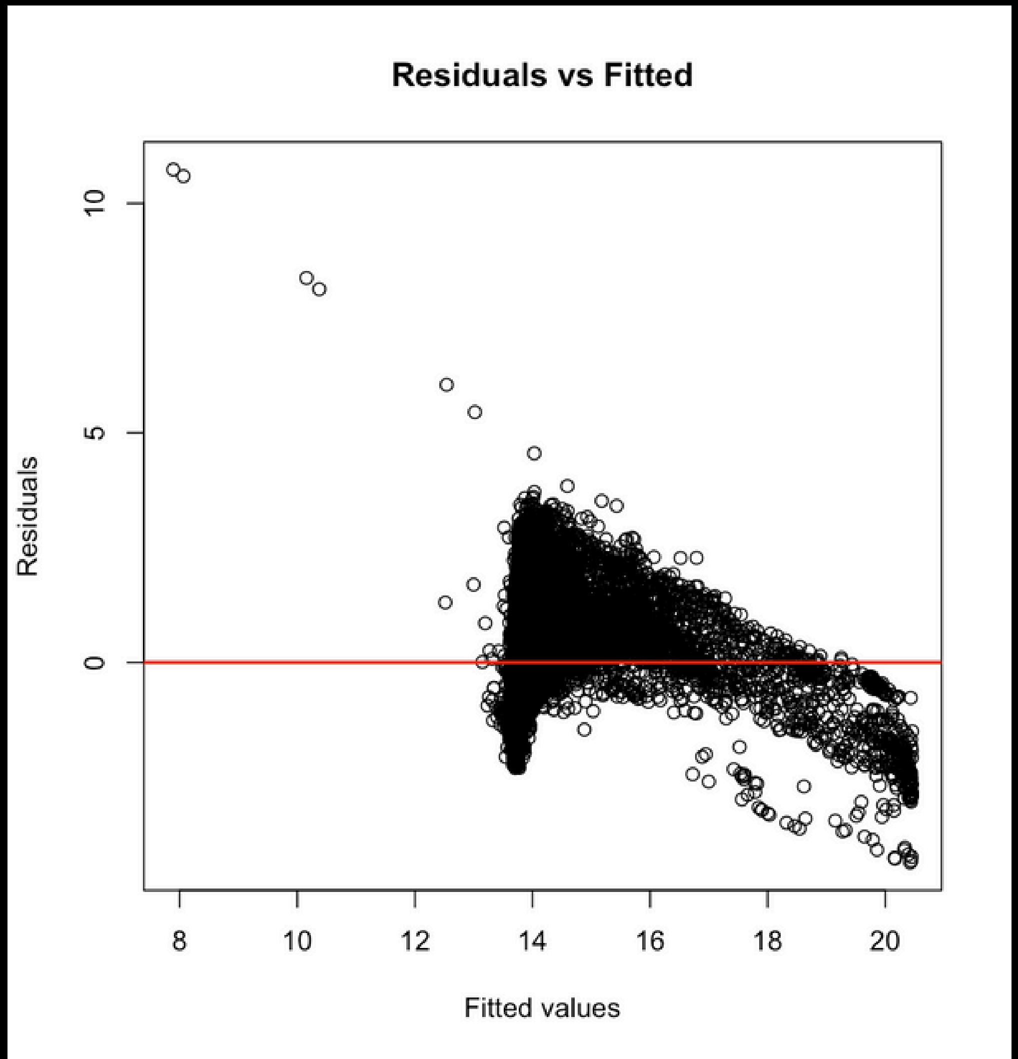
- Final model with excluded insignificant variables and stepAIC to verify any variables that need to be dropped.

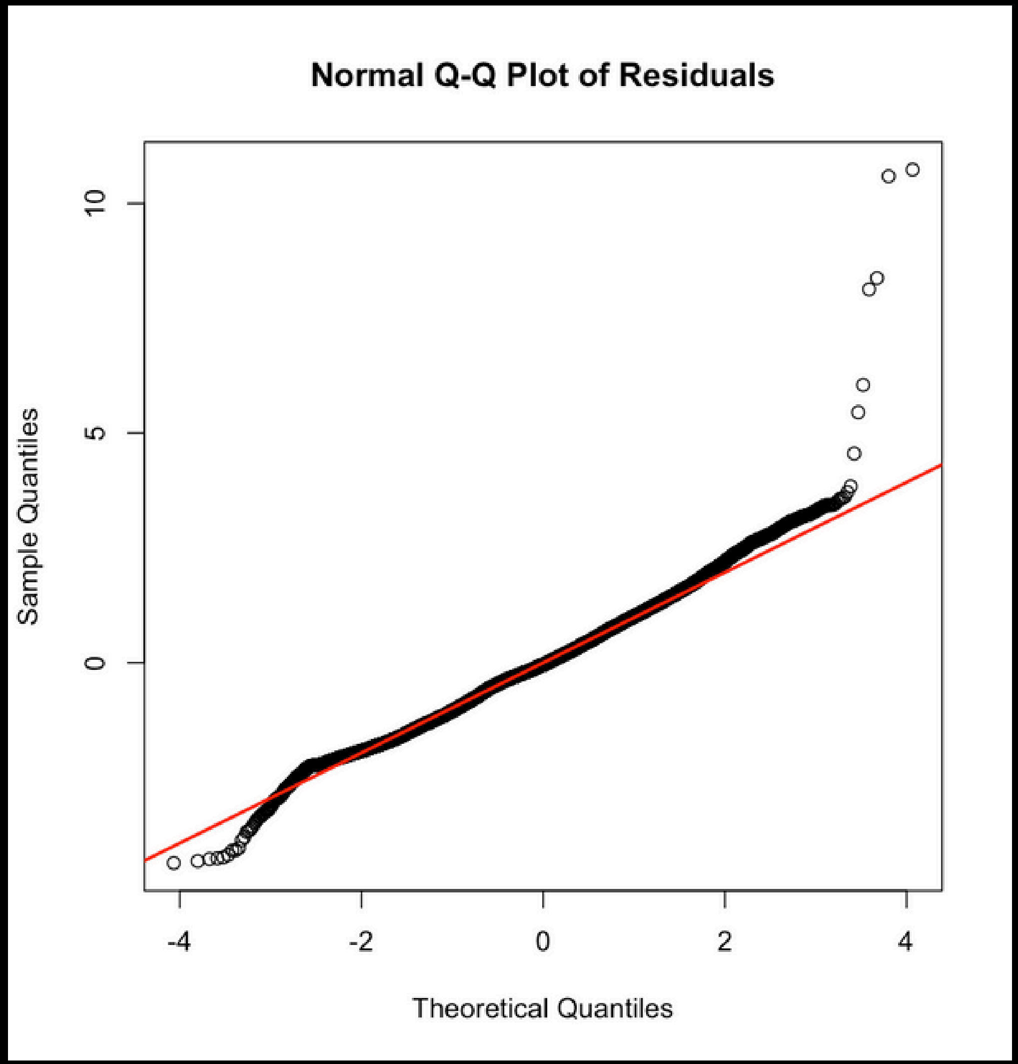
```
Call:
lm(formula = log(Hours_watched + 1) ~ poly(Hours_streamed, 2) +
    poly(Avg_viewer_ratio, 2)[, 1], data = twitch)

Residuals:
    Min      1Q  Median      3Q     Max
-4.3630 -0.6628 -0.0439  0.6620 10.7355

Coefficients:
                                Estimate Std. Error  t value Pr(>|t|)
(Intercept)                    14.147681   0.007117 1987.966   <2e-16 ***
poly(Hours_streamed, 2)1      123.629585   1.032021  119.794   <2e-16 ***
poly(Hours_streamed, 2)2      -78.856981   1.031895  -76.420   <2e-16 ***
poly(Avg_viewer_ratio, 2)[, 1]  -2.361497   1.032613   -2.287   0.0222 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.031 on 20996 degrees of freedom
Multiple R-squared:  0.491,	Adjusted R-squared:  0.4909
F-statistic:  6752 on 3 and 20996 DF,  p-value: < 2.2e-16

Start:  AIC=1299.72
log(Hours_watched + 1) ~ poly(Hours_streamed, 2) + poly(Avg_viewer_ratio,
    2)

                            Df Sum of Sq   RSS     AIC
<none>                                   22330  1299.7
- poly(Avg_viewer_ratio, 2)  2       6.4 22337  1301.8
- poly(Hours_streamed, 2)    2   21388.6 43719 15404.4
```

RESIDUAL ANALYSIS

- The residuals of this model are nearly normally distributed which can be shown in the shape of the histogram and the QQ plot. The residuals are randomly dispersed around 0.







K-FOLD CROSS-VALIDATION

- We then repeat the process to validate the effectiveness of the model on future data.
- The cross validation shows that the model does not overfit or underfit and ~49% of the data is accounted for after training with 9 folds of the data and testing on one fold of data.

```
Linear Regression

21000 samples
    2 predictor

No pre-processing
Resampling: Cross-Validated (10 fold)
Summary of sample sizes: 18900, 18900, 18900, 18900, 18900, 18900, ...
Resampling results:

  RMSE      Rsquared   MAE
  1.032457  0.4905036  0.8041308

Tuning parameter 'intercept' was held constant at a value of TRUE
```

RIDGE & LASSO

- To prevent overfitting and to verify if any more terms need to be dropped, ridge and lasso is performed.
- This is effective especially since there are polynomial terms in our model.
- Ridge and Lasso will shrink the coefficients to nearly zero and eliminate or stabilize terms if needed.

RIDGE & LASSO

Model Summary

```
Coefficients:
                               Estimate Std. Error
(Intercept)                   14.147681   0.007117
poly(Hours_streamed, 2)1     123.629585   1.032021
poly(Hours_streamed, 2)2     -78.856981   1.031895
poly(Avg_viewer_ratio, 2)[, 1] -2.361497   1.032613
```

Ridge Coefficients

lambda.min = 0.08537327

```
4 x 1 sparse Matrix of class "dgCMatrix"
                    s=0.08537327
(Intercept)     14.147681
h1             116.725784
h2             -74.451000
v1              -2.613704
```

Lasso Coefficients

lambda.min = 0.003871574

```
4 x 1 sparse Matrix of class "dgCMatrix"
                  s=0.003871574
(Intercept)      14.147681
h1              123.088024
h2              -78.313632
v1               -1.839042
```

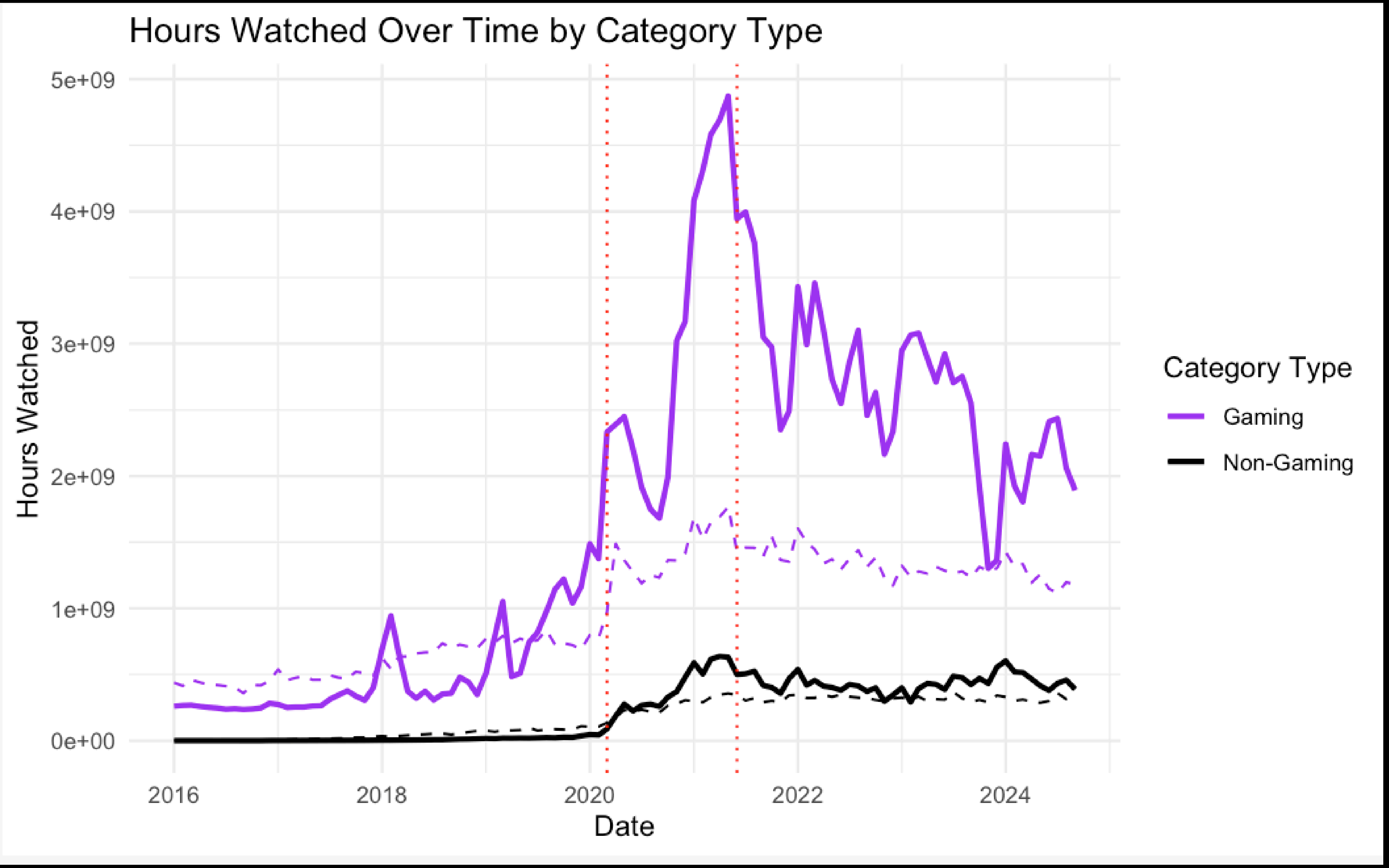
RIDGE & LASSO

- In both the ridge and lasso models, the lambda value is small. Ridge made slight stabilization adjustments to remedy slight multicollinearity in the polynomial terms. Lasso deems the model strong and not overfitting since the coefficients were close to OLS but not identical.
- The Ridge and Lasso technique verifies that all variables in this model are to be included and are all great contributors to the prediction of hours watched on Twitch.
- The final model is:

$$\log(\text{Hours\_watched} + 1) = 14.834 + 0.001573 \cdot \text{Hours\_streamed} - 1.32 \times 10^{-8} \cdot \text{Hours\_streamed}^2 + 0.00421 \cdot \text{Avg\_viewer\_ratio}$$

APPLYING THE MODEL

- In this model, the dashed lines are the predicted values and the solid lines are the observed values.
- Using the finalized regression model, the hours watched for gaming categories and non-gaming categories can be predicted.
- Gaming still remains the most prominent type of content.
- There is an increase in non-gaming content during the pandemic.



CONCLUSION

- The streaming landscape expanded and diversified during the pandemic.
- The pandemic fueled a rapid increase in viewership and content.
- Categories such as Just Chatting, ASMR, Music, Art, Food & Drink, etc. became more popular as a result of the demand in community, entertainment, and distraction from the hardships of the pandemic.
- Games will always be the primary content on Twitch and the most popular games will circulate according to the demands of society.
    - Ex: Accessible multiplayer games like Among Us or free-to-play FPS games like Fortnite and Valorant being popular during lockdown.

CONCLUSION

- The final regression model declares the significance hours streamed and the average viewer ratio plays in hours watched on Twitch. The more content that is streamed on Twitch, the more there is for viewers to watch.
- This model can be used to predict trends in viewership and categories over time through not only the pandemic but other key moments like promotional events, holidays, or product releases.